

HANWEN DENG

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PROFESSIONAL SUMMARY

Postgraduate researcher with over 4 years of hands-on experience across Yale University and University College London, specialising in computational psychiatry, affective neuroscience, and neuroimaging. Contributed to 2 publications, accepted for an oral presentation and 2 poster presentations at conferences. Skilled in R and MATLAB; experienced in computational modelling, ecological momentary assessment (EMA), and fMRI methods applied to mood disorders and addiction. Seeking to advance in a PhD programme in computational psychiatry or clinical neuroscience.

EDUCATION

University College London & Yale University London, UK | Sept 2023 – Sept 2025
MRes Developmental Neuroscience and Psychopathology | Distinction

University of Glasgow Glasgow, UK | Sept 2020 – Sept 2021
MSc Psychological Studies Conversion | Distinction

Southern Medical University Guangzhou, China | Sept 2016 – Jul 2020
BSc Biomedical Engineering | GPA: 3.1/4.0

CONFERENCES & PRESENTATIONS

Oral Presentations

- **When Data Quality Speaks: Computational Markers and Clinical Vulnerability in Early Opioid Use Disorder Treatment.** APA Division 12 (Society of Clinical Psychology) Annual Conference, 2026 — Accepted for Rapid Fire Oral Presentation; unable to attend.

Poster Presentations

- **Whole-Brain Resting-State Functional Connectivity Prediction of Ambiguity Tolerance.** Computational Psychiatry Conference, 2026 — Accepted (in preparation)
- **Discerning Individual Differences in Symptom Trajectories.** YPGA Symposium, 2025 — IMPACT-MH.
 - IMPACT-MH (Individually Measured Phenotypes to Advance Computational Translation in Mental Health) is a large-scale longitudinal study across 16 clinics recruiting N=1,800 clinical and N=600 healthy control participants.
 - Presented preliminary data (N=218; 58.3% female) showing high diagnostic complexity: 70% of participants carried multiple diagnoses spanning schizophrenia/psychosis, depression, PTSD, bipolar disorder, OCD, and substance use disorders.
 - Longitudinal transdiagnostic design combining standardized clinical assessments with computational behavioural tasks and theory-driven modelling to characterise individual symptom trajectories and link intra-individual variability to underlying mechanisms.

HONOURS & AWARDS

National Mathematical Contest in Modeling — Third Prize 2019
Competed among 42,992 teams nationwide across four core problems including multibeam bathymetric data processing, complex logistics optimisation, airport taxi scheduling, and automobile assembly line balancing.

12th Certification Cup Mathematical Modeling Network Challenge — Third Prize 2019
Competed in the national online mathematical modelling challenge. Solved Problem A (Continuous): investigation and optimisation of car-following models on expressways. Competition also included Problem B (Discrete): congestion node identification and scheduling optimisation in traffic networks; and Problem C (Big Data): consumer purchasing behaviour prediction and user profiling based on e-commerce data.

National English Competition for College Students (NECCS) — Third Prize, Category C 2019
Category C is designated for non-English-major undergraduates; tests academic reading, writing, and oral communication.

Challenge Cup Extracurricular Academic Science and Technology Competition — Third Prize 2018
Project title: CT image-based 3D reconstruction in laparoscopic D3 radical resection for rectal cancer.

National College Student's English Reading Contest — Second Prize, Preliminary Competition Foreign Language Teaching and Research Press, 2018

National College Student's Speaking Contest — Third Prize, Preliminary Competition Foreign Language Teaching and Research Press, 2017

"Love Our University" Micro Video Shooting Competition — Second Prize Propaganda Department of SMU, 2016

RESEARCH EXPERIENCE

Neurobiological Markers of Opioid Use Disorder Jan 2026 – Present
Postgraduate Associate, Dr Sarah Yip, Yale University

- Analysed whole-brain functional connectivity in OUD (n=53) vs healthy controls (n=38) across MID reward and Stroop cognitive control tasks using Network-Based Statistics (NBS, 268-node Shen atlas); applied repeated-measures mixed-effects network analysis (NBR, R) and conjunction analysis to establish robustness of findings.
- Identified large-scale FC networks robustly differentiating OUD from healthy controls; demonstrated that group differences persist in resting-state FC, supporting a stable trait interpretation rather than task-driven artefact.
- Found that diagnostic FC networks do not predict treatment response, suggesting a dissociation between neural markers of OUD pathology and markers of recovery, consistent with previous addiction neuroscience literature.

Whole-Brain Resting-State fMRI Prediction of Ambiguity Tolerance Dec 2025 – Present
Postgraduate Associate, Dr Sarah Yip, Yale University & Dr Anna Konova, Rutgers University

- Applied connectome-based predictive modelling (CPM) to resting-state fMRI data (n=93) to predict individual differences in ambiguity tolerance derived from an economic decision-making task.
- Findings suggest that state-dependent processes underlying ambiguity tolerance may require task-based rather than resting-state connectivity paradigms, directly informing future study design.

Computational Markers and Clinical Vulnerability in Early Opioid Use Disorder Treatment Sept 2024 – Jun 2025
MRes Dissertation (Distinction), UCL / Yale University

- Fitted a utility-based computational model to a Risk and Ambiguity Decision-Making Task (n=15, 8-week dense-sampling design) to extract ambiguity tolerance and risk tolerance parameters; tested whether ambiguity tolerance predicted prospective opioid use via time-lagged mixed-effects logistic regression.
- Identified systematic non-random missingness (34.5% of sessions excluded); studied as a clinical signal and applied propensity score weighting to correct for exclusion bias.
- Mediation analysis identified distinct pathways from clinical vulnerability to data dropout: cognitive ability and positive affect were protective, while craving, sleep disturbance, and distress intolerance operated through active substance use.

Mood Velocity and Dynamics in Bipolar Disorder May 2024 – Present
Voluntary Research Assistant, Dr Liam Mason, UCL

- Analysed longitudinal EMA data capturing mood fluctuations in bipolar disorder and healthy control cohorts across 8-week assessment periods.
- Applied continuous-time dynamical systems modelling, including Damped Linear Oscillator (DLO) modelling and Generalised Local Linear Approximation (GLLA), to characterise mood instability parameters.
- Co-authoring manuscript on mood velocity metrics as prospective indices of manic symptom intensification.

Cross-Country Analysis of Perceived Coercion During COVID-19 Aug 2024 – Mar 2025
Voluntary Research Assistant, Dr Veronica Ranieri & Dr Rob Saunders, UCL

- Led multilevel factor modelling and meta-analysis examining perceived coercion across 11 countries; investigated links with anxiety, depression, and emotion regulation.
- Identified significant cross-country variation in coercion scores, with perceived coercion consistently associated with elevated depression, anxiety, and stress symptoms.

PROFESSIONAL EXPERIENCE

Yale University, Department of Psychiatry — IMPACT-MH Program

New Haven, CT | Jul 2025 – Present

Postgraduate Associate (Dr Chris Pittenger & Dr Sarah Yip)

- Conduct structured clinical interviews using the DIAMOND protocol (DSM-5-based) across schizophrenia-spectrum, mood, and personality disorder populations.

St Andrew's Healthcare — Neuropsychiatry Division

Northampton, UK | Nov 2022 – May 2023

Assistant Psychologist

- Provided specialist psychological support for patients with traumatic brain injuries and complex comorbidities (adult women) in a secure neuropsychiatric setting.
- Administered and interpreted neuropsychological assessments (ACE-III, BATCH, WASI-II, RBANS) to inform individualised treatment planning.
- Authored clinical formulations and CPA reports for MDT care planning; delivered psychoeducation to healthcare professionals on neuropsychological approaches to challenging behaviour.
- Developed evidence-based Positive Behaviour Support (PBS) plans incorporating behavioural recordings and person-centred approaches.

St Andrew's Healthcare — Low Secure Rehabilitation

Northampton, UK | Dec 2021 – Nov 2022

Dialectical Behaviour Therapy (DBT) Assistant

- Borderline personality disorder population, age 18-24
- Led daily mindfulness and goal-setting sessions; led weekly homework groups and DBT skills training workshops.
- Performed post-crisis debriefing sessions using Behavioural Chain Analysis to support reflective learning.

Southern Medical University — Student Union, General Affairs Department

Guangzhou, China | Sept 2016 – Jul 2018

Minister of General Affairs

- Planned and organized large-scale school activities, secured funding of 2,000 RMB for an activity with a total of 240 participants.
- Addressed student concerns regarding accommodation, healthcare, shopping, and communication; launched health assessment events to promote campus life quality.

PUBLICATIONS

- **Deng, H.**, Mason, L. (in preparation). Negative affect damping as a prospective index of manic symptom intensification in bipolar disorder: a longitudinal ecological momentary assessment study.
- Liu, M.*, Wu, K., **Deng, H.**, & Huang, X. (2019). Auto insurance renewal forecast and scheme optimization based on customers' characteristics. *Han Journal of Data Mining*, 9(3), 69–80.
- Chen, J., Chen, Z., Guo, W., Xiao, Q., **Deng, H.**, et al. (2019). CT image-based 3D reconstruction in laparoscopic D3 radical resection for rectal cancer. *Chinese Journal of Medical Physics*, 36(7), 778–783.

TECHNICAL SKILLS

Programming & Software

R, MATLAB, NVivo, PsychoPy, Mimics, SolidWorks, CAD

Neuroimaging & Analysis Methods:

Connectome-Based Predictive Modelling (CPM); Network-Based Statistics (NBS);

Statistical & Computational Models:

Linear Mixed-Effects Models (LME); Generalised Linear Mixed Models (GLMM); Multilevel Logistic Regression; Continuous-Time State-Space Model--Damped Linear Oscillator (DLO) Model; Generalized Local Linear Approximation (GLLA); Reinforcement Learning (RL) Model; Expected Utility Model; Prospect Theory Model; Softmax/Luce Choice Model; propensity score weighting, multilevel factor modelling, permutation testing.